

Sewon Park

AI-native Platform & Product Engineering Lead

Engineer with roughly eight years of experience connecting physical devices, enterprise software, developer platforms, and production-grade AI verification. I work where device packets, APIs, agent workflows, and release confidence meet.

FOCUS	DIFFERENTIATOR	OPERATING MODEL
AI platform · DevEx · industrial software	Device boundary to developer abstraction	Verification before trust

01 Selected impact

2026 · PRODUCTIZING

Product owner · architecture · full-stack

BioStar Developer Portal with natural-language verification

Independently designed and implemented an integrated Developer Portal in eight weeks using an AI-native workflow. Documentation, an API source of truth, a device emulator, MCP tools, a Linux/AWS sandbox, and evaluation gates share one operation model. Complex user, door, authentication, and event histories can be recalled and replayed in natural language for immediate verification.

- One operation model shared by six developer-facing surfaces
- Same gateway concept from device emulator toward physical-device workflows
- Recognition through a company-wide engineering excellence program

2026 · IN PROGRESS

~200K LOC modernization with meaning-preservation gates

Leading the frontend team through an AI-native rewrite of about 200K LOC from a mixed Angular/React architecture. Backend payloads and responses, API contracts, React behavior, useEffect, DOM, UI, CSS assertions, and E2E flows are verified as separate release gates.

2021–2024 · CORE PLATFORM

Gateway, device, and distributed-event platform

Implemented a Spring Cloud Gateway MSA foundation and vertically integrated device packets, identity, authentication, and event flows across C++, Java/Kotlin, and web products. A distributed-event pipeline redesign improved processing performance by 40%+, while targeted data-structure and query work reduced CPU processing by about 50% and bulk deletion from the 30-second range to about three seconds.

2021.02—
PRESENT

Platform & Product Engineering · AI-native Mission Lead

Suprema · Seongnam, Korea

- As Integration Development Team Lead, lead Developer Portal productization, ~200K LOC modernization, and product/version-aware E2E verification as cross-functional AI-native missions.
- Developed vertically across C++ device packets and SDKs, Java/Kotlin services, user and identity domains, distributed events, web products, and cloud integrations, including new-hardware features and cross-layer debugging.
- Designed reusable AI verification workflows, led three interns through LLM-assisted unit-test generation, and shared verification-first practices through internal technical sessions and external mentoring.

2020.04—2021.01

Research Engineer · Sensor Software

HBrain · Korea

- Developed Windows software for radar, camera, and FPGA-based pedestrian-detection systems using C/C++, MFC, OpenGL, and device SDKs.
- Contributed to public R&D projects and wrote requirements, protocol, and operational-scenario documents.

2018.10—2020.02

Software Developer · Network Management

AtData · Korea

- Developed and maintained web applications for SK Telecom's network-management program, including IP configuration, transmission views, and raw-data processing.
 - Implemented Jenkins-based deployment automation and supported new server environments.
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RESEARCH IN PROGRESS

Explainable retrieval & hybrid model evaluation

- Expert-knowledge and VLM-based explainable traditional-dance retrieval and recommendation
- Hybrid quantum–classical framework comparing classical ML and QML candidates under shared data splits, budgets, and evaluation gates

2026—PRESENT

Ph.D. coursework, AI Convergence

Sungkyunkwan University

2019—2021

M.S., Applied Data Science

Sungkyunkwan University

Thesis: response-target prediction for public petitions under class imbalance

2010—2017

B.S., Maritime Police Systems

Gyeongsang National University

Platform & product

Java · Kotlin · Python · JavaScript/TypeScript · Spring · React · REST · gRPC

Device & domain

C/C++ · device SDKs · packet debugging · biometric authentication · user/door/auth/event flows

AI & evaluation

LLM · MCP · RAG · agent workflows · PyTorch · contract/unit/DOM/UI/CSS/E2E testing

Data & infrastructure

MariaDB · Microsoft SQL Server · AWS Lambda/S3/EC2 · Docker/Linux · Jenkins · GitLab CI/CD

Mentoring & engineering enablement

Mentored AI service projects across Codeit, an AI bootcamp, and SK Networks Family AI Camp cohorts 27 and 28 for roughly one year. Current teams include a traffic-dispute AI service and a personalized fashion-recommendation service. Internally, led three interns and shared verification-first AI-native practices through technical sessions and examples.

Recognition & selected presentation

One of five finalist teams, the AI-native Developer Portal received KRW 3 million in development funding and later won the company-wide excellence award with a KRW 10 million prize. I also received the AI-category Innovation Award encouragement prize for applying AI throughout 2025 to raise organizational productivity and technical competitiveness. Presented AI traffic-safety work at the Korean Society of Intelligent Transport Systems.